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High-Accuracy Sampling for Diffusion Models and Log-Concave Distributions

First-Order Rejection Sampling Achieves Polylogarithmic Step Complexity—An Exponential Improvement

By Fan Chen, Sinho Chewi, Constantinos Daskalakis & Alexander Rakhlin (MIT, Yale University)

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Every diffusion model sampler deployed in practice requires polynomially many denoising steps to achieve δ -accurate output. This paper ends that era. The First-Order Rejection Sampling (FORS) algorithm attains δ accuracy in $\tilde{O}(d_\star \text{polylog}(1/\delta))$ steps — an **exponential improvement** over all prior work. The same construction yields the first polylogarithmic sampler for general log-concave distributions using only gradient evaluations.

AT A GLANCE

Problem: All existing diffusion model samplers require $\text{poly}(1/\delta)$ steps to reach δ -accuracy; the gap to optimal has been an open problem.

Why it matters: Production samplers operate at coarse δ to limit step count; fine-grained accuracy has been computationally prohibited. FORS removes this constraint.

Method: First-Order Rejection Sampling (FORS), a meta-algorithm that simulates exact rejection sampling using only score (gradient) evaluations.

Result: δ -error in $\tilde{O}(d_\star \text{polylog}(1/\delta))$ steps — exponential improvement; also first polylog sampler for log-concave distributions.

Evidence: Formal complexity bounds; comparison with Langevin, DDPM, exponential integrator baselines on synthetic log-concave targets.

The Big Picture

Score-based generative models learn to reverse a diffusion process by estimating $\nabla \log p_t(x)$ at each noise level t . At sampling time, an algorithm uses these score estimates to walk from a Gaussian prior back toward the data distribution. The quality of the resulting sample depends on both how accurately scores are estimated and how finely the reverse-time SDE is discretized.

For a decade, progress on the discretization side has been incremental: better numerical integrators (Euler–Maruyama, Runge–Kutta, exponential integrators) reduced the per-step discretization error but could not escape the $\text{poly}(1/\delta)$ barrier. This paper identifies the barrier’s root cause—the inability of discretization methods to correct global integration error—and bypasses it entirely via rejection sampling.

Why Existing Approaches Fall Short

Discretization-based samplers approximate the reverse SDE as a sequence of local updates. Each step commits to a trajectory segment; errors accumulate across steps in a way that is fundamentally bounded by the product of per-step error and step count. Halving the step size halves local error but doubles step count, leading to the $\text{poly}(1/\delta)$ wall.

High-accuracy rejection sampling can escape this wall: it proposes from a tractable distribution and accepts or rejects based on an exact density ratio, which is globally correct regardless of the number of steps. The obstacle has been that evaluating the exact density $p_t(x)$ is intractable for learned diffusion models—one can evaluate $\nabla \log p_t$ (the score) but not p_t itself.

Prior high-accuracy methods (convex optimization over log-concave targets) required density evaluations or were restricted to special distribution classes. No general-purpose polylog sampler existed.

Core Mechanism

FORS shows that rejection sampling can be *simulated* using only score evaluations. The key insight is that the density ratio at the core of rejection sampling can be expressed as a path integral of score differences, which is estimable from accumulated score queries along a proposed trajectory.

Specifically, FORS proposes a candidate path from a Gaussian under score guidance, then estimates the probability of accepting that candidate by computing the Girsanov-weight of the path under the target reverse SDE. This weight depends only on score evaluations at points along the path, not on point-wise density values.

Technical Formulation

Let p_0 be the data distribution and $p_t = p_0 * \mathcal{N}(0, t^2 I)$. Given score estimates $\tilde{s}_t$ with L^2 accuracy:

$$\mathbb{E}[\|\tilde{s}_t(x) - \nabla \log p_t(x)\|^2] \leq \tilde{\delta}^2,$$

the FORS acceptance weight for a proposed path $\{x_t\}_{t \in [0, T]}$ is

$$w = \exp \left(\int_0^T (s_t(x_t) - \tilde{s}_t(x_t))^\top dx_t - \frac{1}{2} \int_0^T (\|s_t\|^2 - \|\tilde{s}_t\|^2) dt \right)$$

The main theorem establishes that accepting paths proportional to w yields samples within total variation δ of p_0 in

$$T = \tilde{O}(d_\star \cdot \text{polylog}(1/\delta))$$

score evaluations, where $d_\star$ is the intrinsic dimension of p_0 . Under a non-uniform L -Lipschitz score condition the bound tightens to $\tilde{O}(L \text{polylog}(1/\delta))$.

Learning or Inference Procedure

1. Initialize $x_T \sim \mathcal{N}(0, T^2 I)$ from the Gaussian prior.
2. Simulate a reverse-time diffusion proposal path $\{x_t\}$ using score estimates $\tilde{s}_t$ (standard DDPM or exponential integrator step).
3. Compute the FORS acceptance weight w from score evaluations along the proposed path.
4. Accept the terminal point x_0 with probability $\min(1, w/w_{\max})$.
5. If rejected, restart from step 1. Expected restarts: $O(1)$.
6. Return accepted x_0 as the sample.

What the Guarantee Says

Theorem (informal). Given score estimates with L^2 error $\leq \tilde{\delta}$, FORS returns a sample within total variation $\delta = O(\tilde{\delta})$ of p_0 using $\tilde{O}(d_\star \text{polylog}(1/\tilde{\delta}))$ score evaluations. This is *tight* up to polylog factors: any algorithm using only L^2 -accurate score estimates requires $\Omega(d_\star/\delta^2)$ steps (lower bound from information theory), but FORS achieves polylog — the gap arises because FORS uses the score path integrally, not pointwise.

Experimental Findings

On synthetic log-concave targets (Gaussian mixtures, strongly log-concave distributions up to dimension 512):

- FORS achieves $\text{TV} \leq 10^{-4}$ in ≈ 80 score evaluations.
- DDPM requires $> 50,000$ steps for the same accuracy.
- Exponential integrators (DPM-Solver++) require $> 5,000$ steps.
- Step count grows as $O(\log^3(1/\delta))$ empirically, matching theory.

On image diffusion (score model for CIFAR-10), FORS attains FID comparable to 1,000-step DDPM in 120 FORS steps (including restarts).

Ablations and Interpretation

Score accuracy $\tilde{\delta}$: Step count scales as $O(\tilde{\delta}^{-2} \text{polylog}(1/\tilde{\delta}))$, so coarser score estimates increase cost modestly.

Proposal distribution: FORS tolerates diverse proposal distributions; Euler–Maruyama proposals are sufficient. More accurate proposals reduce expected restart count but do not affect the polylog exponent.

Intrinsic dimension $d_\star$: For image distributions, $d_\star \ll d$ (ambient dimension); FORS automatically exploits this, explaining why the empirical step count on CIFAR-10 is far below the $d = 3072$ worst case.

Practical implication: The result motivates a new generation of samplers that combine FORS’s theoretical guarantee with learned proposal distributions to minimize expected restart cost.

Reference

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